A Generalizable Machine-Learning Approach to Mapping Aquaculture Ponds from Satellite Imagery

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Mentors. Alex Wiebe, Postdoctoral Scholar, and Sebastian Heilpern, Assistant Professor, Department of Earth System Science, Stanford Doerr School of Sustainability. Their work connects global change, food systems, biodiversity, and environmental sustainability.

Group meeting. Time and place TBD.

1 Introduction

Aquaculture supplies a growing share of the world's aquatic food and plays an important role in food security and livelihoods. Land-based ponds account for much of this production, but their global spatial distribution remains poorly documented. Without consistent, high-resolution spatial data, researchers cannot reliably assess where aquaculture is expanding or evaluate its relationships with land-use change, biodiversity, water resources, and surrounding ecosystems.

Existing remote-sensing studies provide an important foundation but differ in geographic coverage, temporal extent, and spatial resolution. Wang et al. [3] produced a global map of landside clusters of aquaculture ponds from 2020 Sentinel-2 imagery, while Ottinger et al. [2] mapped individual ponds across the coastal zone of Asia using Sentinel-1 and Sentinel-2 time series. Jiang et al. [1] reconstructed changes in coastal aquaculture ponds in China and Southeast Asia between 1990 and 2020 using Landsat imagery. These studies motivate a workflow that emphasizes pond extent and systematic evaluation across diverse regions.

This project will develop and evaluate a scalable machine-learning workflow for segmenting aquaculture ponds from satellite imagery. The primary goal is to delineate pond extent using pixel-level masks suitable for estimating area and, where feasible, conversion to polygons. Detection may serve as an intermediate step or fallback if reliable segmentation proves infeasible. The quarter's primary outcome will be an independently evaluated proof of concept across several geographically distinct study regions. Broader regional or global mapping will depend on data access, computing resources, and model performance.

2 Research Questions and Deliverables

The project will address two linked research questions:

- How accurately can a satellite-imagery workflow delineate aquaculture pond extent across geographically distinct regions?
- Does multimodal large-language-model post-processing reduce false positives or identify uncertain cases more effectively than unfiltered and rule-based baselines?

The primary deliverables for the quarter are:

- A harmonized training dataset using preliminary labels from existing products, with separate, independently manually validated evaluation data.
- A reproducible baseline segmentation pipeline and an Earth foundation model comparison.
- A geographically held-out evaluation, including an ablation of multimodal post-processing.
- Study-region pond masks, area estimates, and confidence estimates, with reproducible code and documentation; polygon outputs where feasible.

A larger-scale or global map, historical mapping, and classification by production group remain stretch goals rather than required quarter deliverables.

3 Methodology

3.1 Data Collection and Preprocessing

We will assemble existing aquaculture maps and corresponding Sentinel-1 and Sentinel-2 imagery. Candidate sources include the products in [1-3] and the Aquaculture Earth coastal mapping product [4]. We will assess source coverage, dates, spatial resolution, class definitions, and licensing before use. Initial study regions will balance label availability, geographic diversity, and contrasting pond landscapes. Inputs will be harmonized to common coordinate systems, spatial resolutions, and observation windows, with source limitations documented.

3.2 Training Labels and Independent Reference Data

Existing aquaculture products will provide preliminary training labels for pond extent where supported by their resolution and definitions. Cluster-level products will not be assumed to delineate individual ponds. Within valid overlapping coverage, agreement and disagreement will define sampling strata for manual quality checks; agreement alone will not establish correctness. Missing coverage will not be treated as a negative label. Training examples will include hard negatives such as rice fields, salt pans, reservoirs, and natural water bodies.

Separately, we will build an independent reference set in geographically held-out validation and test regions. Human reviewers will delineate or validate pond presence and boundaries from temporally matched imagery, using higher-resolution imagery where available. Reference areas will include both pond and non-pond landscapes and will not be restricted to existing product labels or model detections, allowing missed ponds to be assessed. Reviewers will interpret imagery independently of model predictions and, where feasible, existing product overlays; uncertain cases will be flagged and reviewed. Annotation extent and sample size will be agreed with mentors within the quarter's capacity.

Validation data will support model and threshold selection. The final test reference will be frozen before model comparison and excluded from training, tuning, and model selection. Shared errors across existing products will therefore not serve as the primary standard for final performance.

3.3 Aquaculture Pond Segmentation

We will first establish an interpretable pixel-level classification baseline using spectral and temporal features, with geometric filtering where appropriate. We will then evaluate whether representations from an Earth foundation model improve segmentation using the same splits and metrics. Predicted masks will support pond-area estimation and, where feasible, polygon extraction. Detection may help identify candidate areas or serve as a documented fallback; any fallback will be reported as a limitation relative to the primary segmentation objective.

3.4 Multimodal Post-processing

A multimodal large language model will review a defined subset of predicted pond regions using satellite imagery and relevant geographic context. We will compare no additional filtering, rule-based geometric filtering, and multimodal review on the same candidates. Independent human reference annotations will remain the evaluation standard; LLM outputs will not be treated as ground truth. We will compare precision, recall, segmentation quality, cost, and cases deferred for human review. End-to-end evaluation will also include ponds missed by candidate generation.

3.5 Model Evaluation and Mapping

Evaluation will use geographically separated training, validation, and test regions. Primary metrics will include pond-class intersection over union, pixel-level precision, recall, and F1 score, plus error in total pond area within reference areas. Results will be reported by region and landscape type. We will examine confidence calibration and common failure modes, including uncertain boundaries and temporal mismatches. After selecting the workflow using validation data, we will evaluate it on the frozen test set and generate study-region maps. Expansion will depend on imagery, computing budget, and observed generalization.

4 Schedule and Milestones

Weeks	Focus	Expected output
1	Scope and setup	Confirm study regions, datasets, computing constraints, segmentation target, and independent evaluation protocol with mentors.
2-3	Data preparation	Harmonize imagery and preliminary training labels; document source coverage; sample independent reference areas and begin manual annotation.
4	Label and reference review	Complete stratified training-label checks and reference annotation; review uncertain cases; freeze geographic splits and the independent test reference.
5-6	Baseline modeling	Train the conventional segmentation baseline; assess development-validation results and failure modes.
7	Foundation model	Evaluate an Earth foundation model using the same training and validation splits and segmentation metrics.
8	Post-processing comparison	Compare unfiltered, rule-based, and multimodal outputs on a common candidate set using validation data; select the final workflow.
9	Mapping and final evaluation	Evaluate the selected workflow and prespecified ablations on the frozen test set; generate study-region maps, area estimates, and failure-mode summaries.
10	Final delivery	Complete the report, reproducible pipeline, presentation, and recommendations for scaling.

References

[1] Yufeng Jiang, Jing Li, Zhen Zhang, and Yi Li. Dynamics of coastal land-based aquaculture pond in China and Southeast Asia from 1990 to 2020. *International Journal of Applied Earth Observation and Geoinformation*, 127:103654, 2024.

[2] Marco Ottinger, Felix Bachofer, Juliane Huth, and Claudia Kuenzer. Mapping aquaculture ponds for the coastal zone of Asia with Sentinel-1 and Sentinel-2 time series. *Remote Sensing*, 14(1):153, 2022.

[3] Zhihua Wang, Junyao Zhang, Xiaomei Yang, Chong Huang, Fenzhen Su, Xiaoliang Liu, Yueming Liu, and Yuanzhi Zhang. Global mapping of the landside clustering of aquaculture ponds from dense time-series 10 m Sentinel-2 images on Google Earth Engine. *International Journal of Applied Earth Observation and Geoinformation*, 115:103100, 2022.

[4] Clark Labs. Aquaculture Earth: coastal pond aquaculture mapping viewer. https://www.aquaculture.earth/coastal/index.html. Accessed September 23, 2026.